

Data Cave Cheatsheet

This is designed to provide step by step instructions for taking a data set (here a CHEF dump) and volume rendering it in the cave. The vGeo getting started document, provides helpful information for the first time user and can be read by typing

evince filename.pdf

The various documentation files are located in

/usr/local/vGeo5/vGeo_5.0RC3/doc

1) Translate a CHEF data set dump into hdf format. This is something that vGeo will accept.

a) On the Mahone cluster, get into the directory with the chef data set. This directory must also have the appropriate chfset file to read the number of zones in each of the three directions as well as the spacing in the θ and ϕ directions.

b) Run the code chef2hdf.exe. This code was produced by Chris Geroux, and the executable is in /home/bdeupree/bin. One can see the options by typing

```
/home/bdeupree/bin/chef2hdf.exe -h
```

For CHEF calculations, one will probably want to scale the coordinates between zero and one because r is in cm (on the order of $10^{*}8$ cm) and θ and ϕ are in radians or degrees. Thus, not scaling the coordinates will produce a long, very skinny line. The command one will probably want will be

```
/home/bdeupree/bin/chef2hdf.exe -r -t -p CHFMOD CHFSET
```

CHFMOD and CHFSET are the chfmod and chfset files you want to use, respectively. The $-r$ refers to scaling the radial coordinate, $-t$ for the theta coordinate, and $-p$ for the phi coordinate. The output file, out.hdf, currently contains the density, temperature, fractional

variation of each of those from the horizontal average, radial velocity, theta velocity, phi velocity, and magnitude of the total velocity.

c) Rename the out.hdf file to whatever you wish, but maintain the .hdf suffix. You now have a file that the vGeo preparation software can read.

d) Alternatively, a more recent version of chef2hdf allows you to utilize files vGeo requires for animation. You create an ascii file (say, bob_file.list) with the names of the data files (with hdf extension) you want to have in the animation. Then you run chef2hdf with the following option:

```
/home/bdeupree/bin/chef2hdf.exe -r -t -p -b bob_file.list CHFSET
```

Where CHFSET is the usual chfset input file. This will go through the files in your list. If the file already has an hdf file made, it will read the time and save it to an array. If not it will still save the time to an array, but will also create the hdf file. The time file written out at the end will be called time.hdf

2) First step in creating a .vgeo file from your .hdf file is to run a Mechdyne code called “hdfConfigWizard”

a) log on to one of the two RedHat 5 Linux workstations in the Graphics Workroom. Bring up a window and switch to the directory that has the .hdf data set in it.

b) If necessary, determine what shell you are using by typing

```
echo $SHELL
```

I am currently using the c shell, so I would source the settings file by typing

```
source /usr/local/vGeo5/setupVGEO.csh
```

For those running the bash shell, replace .csh by .sh

c) Bring up another window (it will be in the /home/bdeupree directory), and type

```
cp vgeorc_desktop .vgeorc
```

d) Go back to the first window (the one in the directory with the .hdf files in it), and run the configuration wizard by typing

```
hdfConfigWizard
```

e) This will bring up a GUI to which you respond with the following information. There are two possible responses, depending on whether or not you are setting up an animation or just a time independent result. Be aware that you can stop an animation at a specific step and play with it just like you could if it was time independent. We first provide the steps for a time independent result:

- i) Select “Create new file” and click “next”
- ii) Select “Gridded data” and click “next”
- iii) Select “No, the data is time independent” and click “next”
- iv) Select “3D” and “Browse”
- v) Select your hdf file from the list and click “open”
- vi) No choice is required, click “next”
- vii) Select “fakeDim0” for the X variable, and click “next”
- viii) Select “fakeDim1” for the Y variable, and click “next”
- ix) Select “fakeDim2” for the Z variable, and click “finish”
- x) Enter a name for the .hfg output file, click “save” and then “OK”

Now for an animation set up:

Before running hdfConfigWizard, create a file which has the names of the individual hdf files in it. There should be one file per time step, and it needs to match the number of time steps you will be entering in when you run hdfConfigWizard.

- i) Select “Create new file” and click “next”
- ii) Select “Gridded data” and click “next”
- iii) Select “Yes, data changes with time” and “Time sequence”
- iv) Here you have two choices: 1) select “Time list” and enter the number of time steps on the appropriate line and the individual time values on the following line **or** 2) select “Time List” and pick “Data Variable” and enter the hdf file name with the time step information with it (e.g., time.hdf from chef2hdf). Then you will need to select “Data-Set-2” from the variable list.

The latter has the virtue that the only things you must do if you add data files are to 1) edit the list with the .hdf filenames, 2) rerun chef2hdf to rewrite the time.hdf file and make an hdf file with the new data dump information in it, and 3) edit a line in the .hfg file to reflect that the number of time entries has been increased by one. This is done by finding the line

```
DIM I () J () K () T ()
```

where each of the parentheses is meant to represent a number (i.e., the parentheses are not present). These contain the dimensions of the calculation in the x, y, and z coordinates and the number of time steps you have in the animation. Change the number after the “T” to be one larger. There remains a potential problem if the dataset you have added has expanded either the maximum or minimum values of any of the variables you have defined (as in “Data-Set-n” or the time). These values appear in the TIME_DEF and VAR_DEF lines of the .vgeo file you will be creating in the next step. The simplest way to make sure your addition catches everything is to rerun the configCreateUI (see step 3 below) utility and set up all the variables you wish to examine just as you did in making the original prototype .vgeo file. You could then cut and paste all TIME_DEF and all the VAR_DEF lines into the .vgeo file you have already created and modified to do what you want to do. If you do not make the changes and the data or time range has increased, you will either have holes in your data or it will be pegged to the maximum or minimum value you currently have (see, for example, the options for the VAR_DEF command on page 89 of the vGeo 5.0 Reference Manual).

v) Select “3D” and “Browse”, and then enter the name of the file which has the list of hdf files in it (I do it one per line). This filename should not have an .hdf extension because it is merely an ascii file.

vi) Select the data files you wish to include, check “list” and then select “Next”

vii) Select “fakeDim0” for the X variable, and click “next”

viii) Select “fakeDim1” for the Y variable, and click “next”

ix) Select “fakeDim2” for the Z variable, and click “finish”

x) Enter a name for the .hfg output file, click “save” and then “OK”

f) This will exit the GUI.

3) We are now ready to run the code which will create the .vgeo file which the cave software vGeo will run.

a) start this code by typing

```
configCreateUI
```

b) When the GUI comes up, select “Non-geospatial”, and then select “NG_RECTANGULAR”

c) switch to the “Data Source” tab and click “Add”

d) Click “OK”

e) Select the .hdg file you created in the hdfConfigWizard step and click “Open”

f) Click “Create Config”, enter a name for your .vgeo output file (it will append the .vgeo), click “Save”, and then click “OK”

g) Click “Exit”.

4) This will have produced a .vgeo file with the name you selected, but currently this has very little in it. The chief thing it does have is information about the coordinates (COORD_DEF), time (TIME_DEF), and the individual data quantities (density, temperature,...) (VAR_DEF), but it only knows them as “Data-Set-n”, where n is an integer. The file also identifies the name of the .hdf file (or the file with the list of the hdf files in it) you are using (DATA_DEF).

5) You can now run vGeo in desktop mode to generate the graphical objects (GOBs) you want. Note that once you generate these, you do not have to regenerate them. You will save a new .vgeo file at the end of the process and all you will have to do is to use that file the next time you start vGeo. You will also need to learn how to navigate your way around the display in desktop mode. The commands are summarized in tables 1-1 and 1-2 on page 5 of the vGeo Quick Start & Demo Guide. I would suggest it might be useful to print a copy of this page off and keep it with you when you come to work on the desktop. Generating new GOBs is for the most part fairly easy and intuitive with the GUI, but the volume rendering is trickier because of the transfer function, so I will indicate the steps here.

The first step is to load the current .vgeo file. Before you can create your GOBs for volume rendering, you must load in the appropriate transfer functions you want to use. These can be created through the GUI based

transfer function creator (tfeditor) or copied from other sources. It is most convenient if they have a .tf extension. Once you have identified the transfer function(s), you can create volume rendering jobs, associating them with one of the transfer functions. You can create and bring in other transform functions later and associate them with particular GOBs as needed.

- a) Select Builders -> Defs -> Transfer Function file. Select a name for the transfer function and choose the appropriate transfer function file. Then load the transfer function and close the window.
 - b) Select Builders -> GOBs -> Volume renderer. Select a name for the GOB (this should be something that makes sense to you, like “density”), and then identify that with one of the data sets associated with the hdf files you wrote (for example, “Data_set_3”). Thus, you have to know what each of the data sets you wrote in the hdf file are. Finally, pick the transfer function you wish to use with that volume rendering.
- 6) Once this is done, you are ready to run vGeo. While you could do this in the cave, I find it helpful to find the nuisance problems at the workstations. Running the cave in desktop mode will not provide the immersive environment, but it will allow you to create new GOBs and see that things are the way you want without having to bring up the cave.
- 7) We did note above what you had to do to add a new dataset to an animation, but did not say what you did not have to do. In particular, you do not have to run either hdfConfigWizard or configCreateUI again. However, do not change the names of any of the files which include the times, the names of the .hdf files for the animation, of the .hfg file. The first two are referenced by the .hfg file and the .hfg file is referenced by the .vgeo file. It is OK to change the name of the .vgeo file itself.
- 8) Once you have set things up properly, you are ready to run in the cave.
- a) start up the power on the small console on the right hand side of the desk – start by waving your hand at it, and then tap the Power On icon in the upper right hand corner. You should head the projector and the fans come on

b) turn on all four monitors (three on the shelf and one on the left hand side of the desk) – If the monitor corresponding to the master (the one on the left hand side of the desk) does not come on, see Phil

c) log on to the master (computer). This is the monitor on the desk

d) in your home directory, bring up a terminal and make sure you are in “cave mode” by copying the cave version of the setup file to the setup file by typing

```
cp vgeorc_cave .vgeorc
```

e) type

```
source /usr/local/lib/cave.csh  
trackd -status
```

f) bring up another terminal (call it terminal 2) and type

```
source /usr/local/lib/cave.csh  
source /usr/local/vGeo5/setupVGEO.csh
```

g) bring up one terminal window on the master computer for each of the four drones and ssh to each drone (e.g., ssh drone1). This will allow the drones to know your directory is accessible

h) in terminal 2, launch cluster launcher by typing

```
ClusterLauncher
```

i) When cluster launcher brings up a window, it wants to know what application you want (vGeo5.0), where to run from (the directory which contains the .vgeo you wish to run), and an “option”. This last really means the name of the .vgeo file you want to bring up.

NOTE: You can run from any directory in your space. However, the vGeo software will not be able to write any files to your directory unless you allow it to by setting your privileges appropriately. This you may be reluctant to do because it will give others write access to your directory. Phil plans to set up directories which individual users will own but which vGeo can write to. You would thus save any changes you want to make to this directory and then move it to your more secure directory later.

When you have finished in the cave, do the following:

- 1) exit ClusterLauncher
- 2) get into the window in which trackd is running and do control-C
- 3) log out of the master computer
- 4) turn off all four computer monitors
- 5) turn off the power on the console on the right hand side of the desk (if the cave does not go dark, inform Phil)
- 6) turn out lights